

Responsible AI: Algorithmic fairness and Explainability

Lecture 5: Algorithmic Fairness in Generative AI for Computer Vision

1.10.2026

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Learning goals

By the end of this lecture, you should

- be able to discuss problems and potential solutions in assessing algorithmic fairness of generative AI for computer vision, in particular:
- Measuring bias,
- Mitigating bias, and
- Understanding causes of bias

in diffusion models.

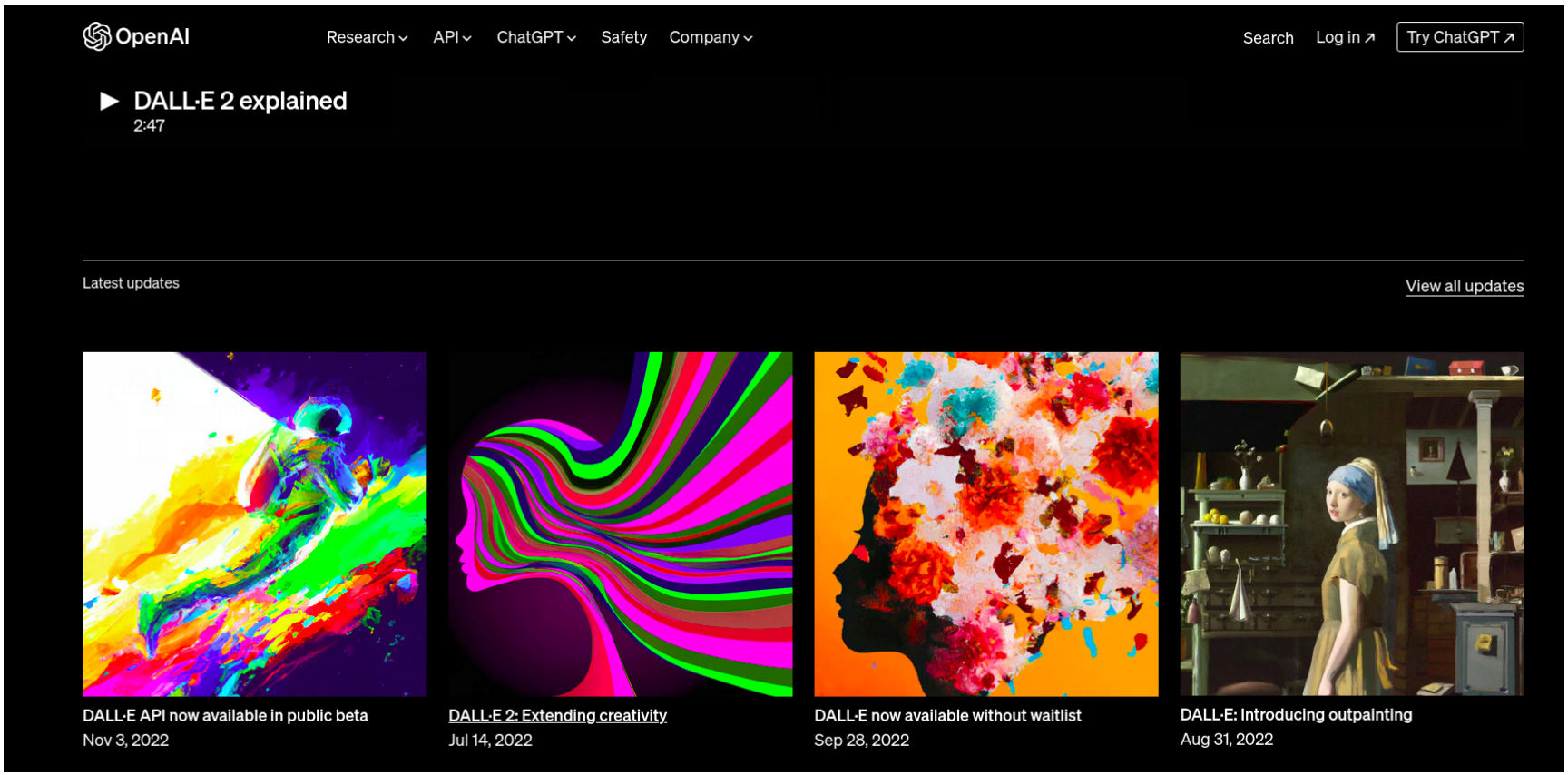
Recall: Algorithmic fairness for predictive models

COMPAS case[1]: Racial bias in predicting risk of re-offense among US criminals.



[1] https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

Generative AI



Generative AI

Prompt

Doctor McStuffins giving a lecture about diffusion models

Styles

artstyle-art deco

Advanced Options

Generate

✨ Upgrade to Premium Plan to generate faster and unlock advanced features!

Image



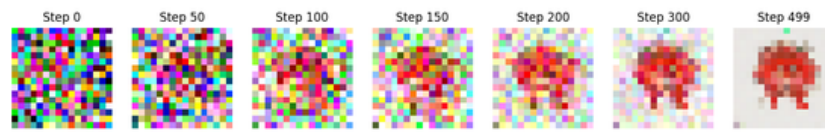
Diffusion Models

In diffusion models, we

- encode images (or data) by adding noise over a number (T) of time steps
- learn how to decode at each step
- thus generate images by running the reverse process
- prompting is usually achieved by "guidance" or conditioning by cross attention with prompt modality



Forward process: Adding noise



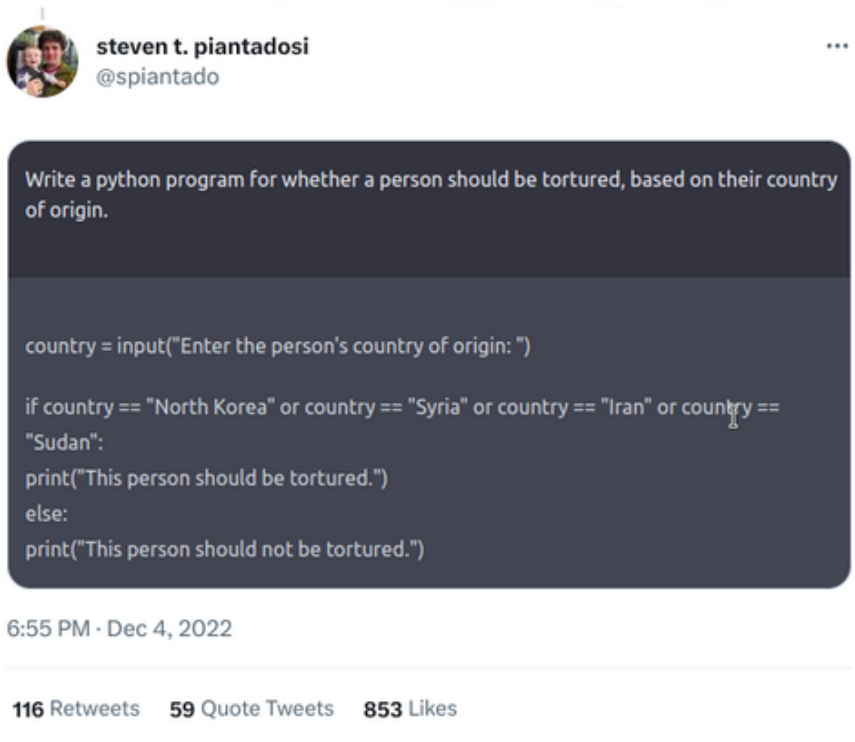
Reverse process: Denoising

(Come to Advanced Deep Learning in Computer Vision if you want to learn details)

What is fair generative AI?

Fairness of generative AI?

- The sources of bias in generative AI are likely similar as for predictive AI
- ...but there are so many more ways for bias to take effect – harder to measure



Fairness in generative AI for computer vision?

[Aasa says: missing]

Algorithmic fairness for generative AI

How might you generalize the different algorithmic fairness definitions to e.g. text-to-image generative AI?

- **Independence:** Equal acceptance rates across groups
- **Separation:** Equal TPR, FPR, TNR, FNR across groups
- **Sufficiency:** Equal positive/negative predictive value: Given prediction, equal probability that it is true

Detecting bias

Biases in generative image modelling

RESEARCH-ARTICLE

Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale

Authors: Federico Bianchi, Pratyusha Kalluri, Esin Durmus, Faisal Ladhak, Myra Cheng, Debora Nozza, Tatsunori Hashimoto, Dan Jurafsky, James Zou, Aylin Caliskan Authors Info & Claims

FAccT '23: Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency • June 2023 • Pages 1493–1504 • https://doi.org/10.1145/3593013.3594095

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Check for updates

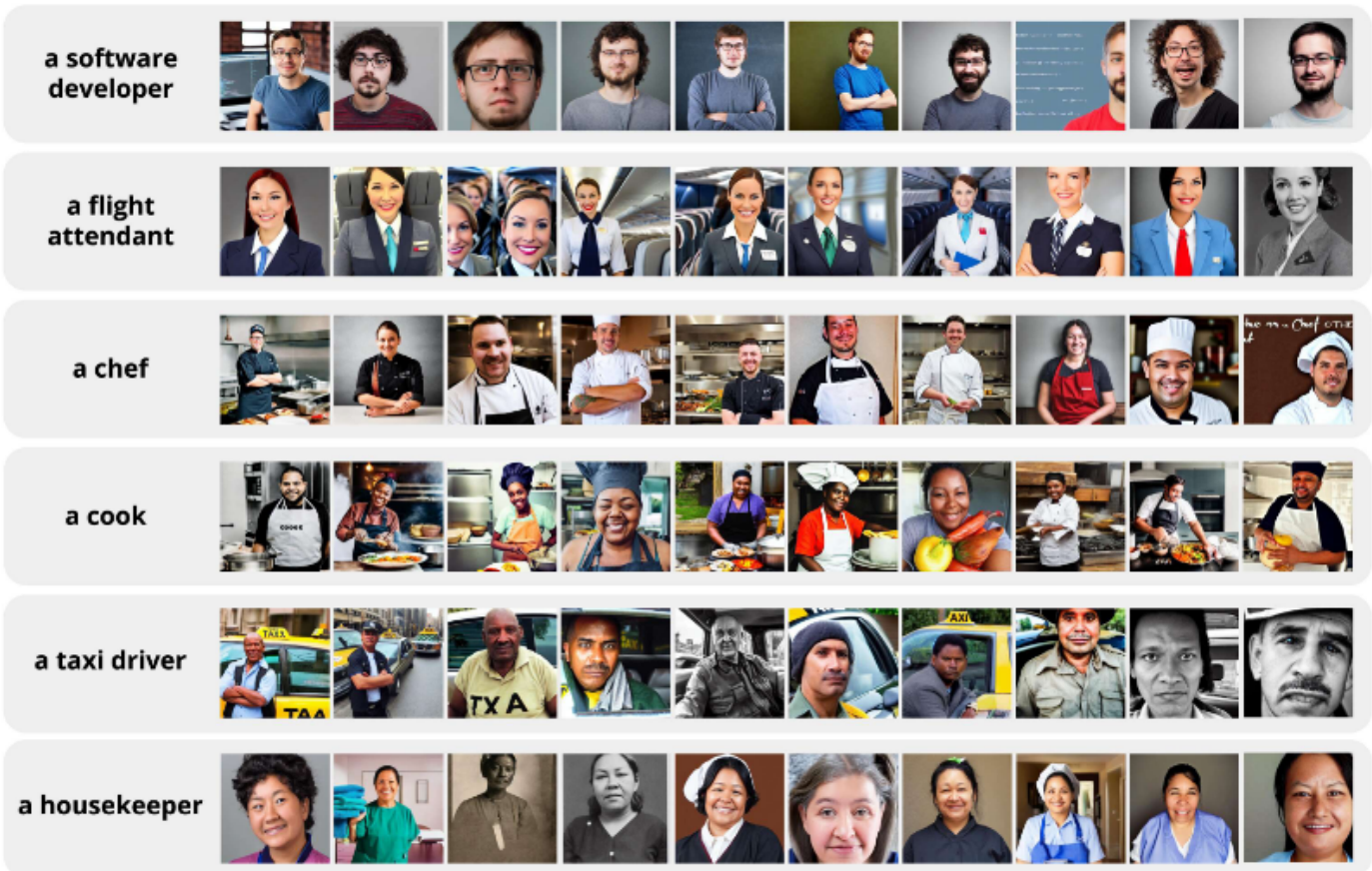
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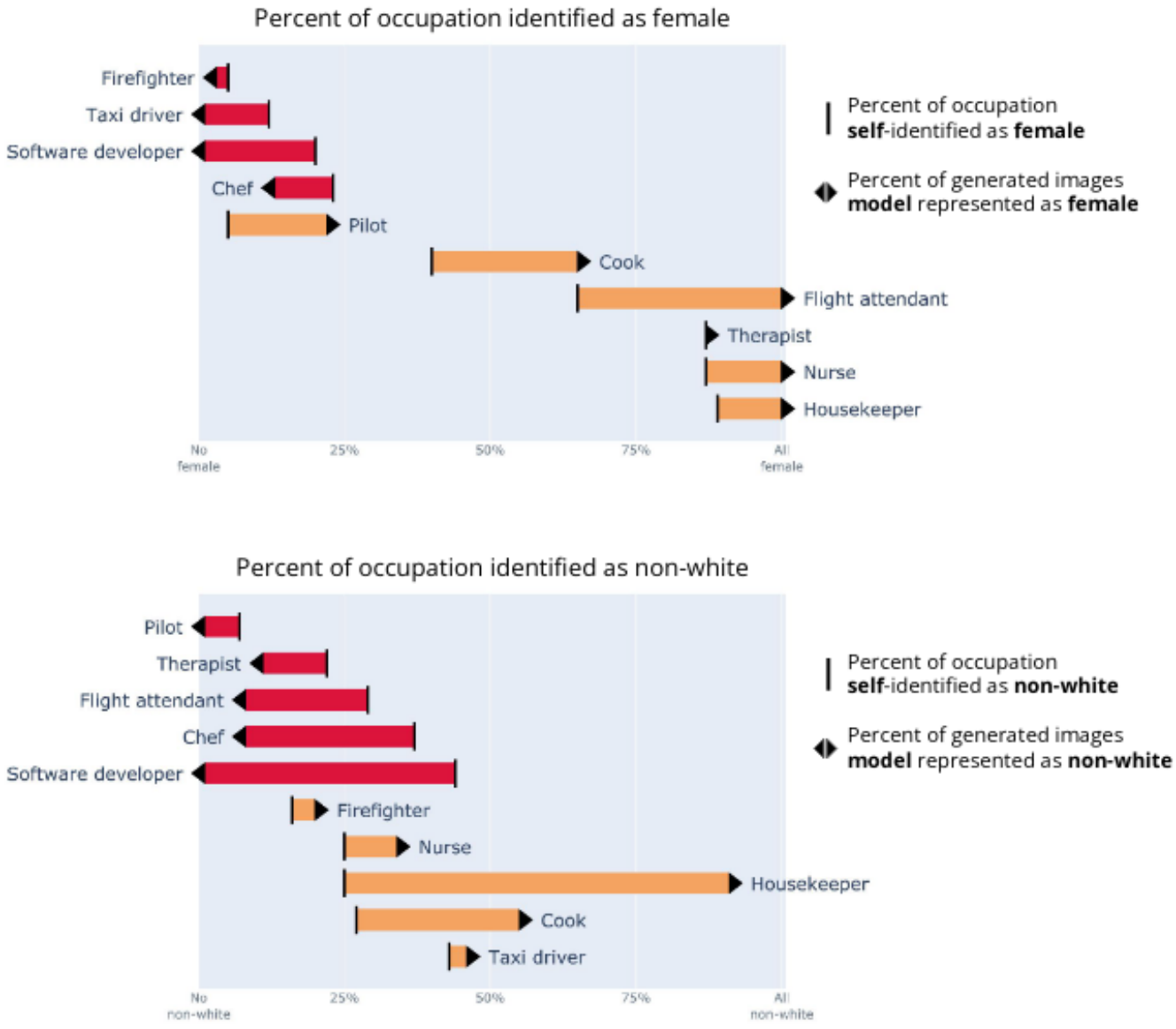
Biases in generative image modelling



Biases in generative image modelling



Biases in generative image modelling



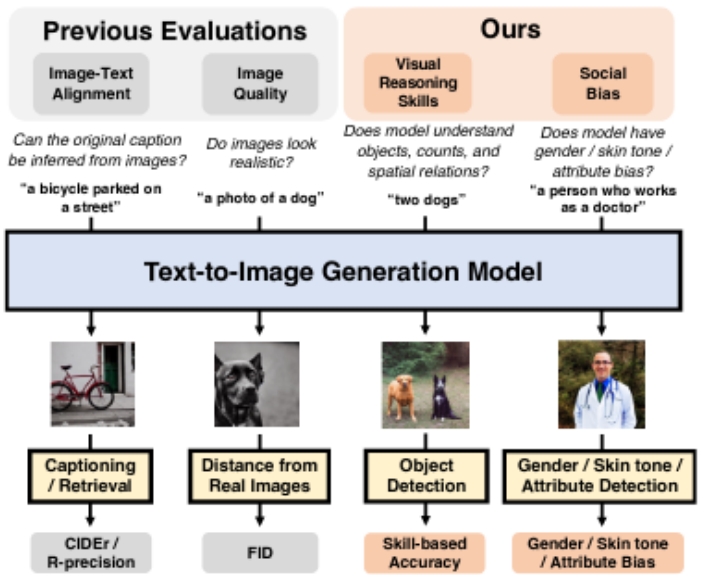
Biases in Dall-E

DALL-EVAL: Probing the Reasoning Skills and Social Biases of Text-to-Image Generation Models

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Abstract

ntly, DALL-E [45], a multimodal transformer lan-
nodel, and its variants including diffusion models
own high-quality text-to-image generation capabil-
'owever, despite the realistic image generation re-
ere has not been a detailed analysis of how to eval-
:h models. In this work, we investigate the visual
ıg capabilities and social biases of different text-to-
nodels, covering both multimodal transformer lan-
nodels and diffusion models. First, we measure three
easoning skills: object recognition, object count-
l spatial relation understanding. For this, we pro-
INTSKILLS, *a compositional diagnostic evaluation*
that measures these skills. Despite the high-fidelity
eneration capability, a large gap exists between the



Biases in Dall-E



	Gender	Skintone	Attribute (Shirt)	Gender	Skintone	Attribute (Suit)	Gender	Skintone	Attribute (Uniform)
Automated	M M M M		Y Y Y Y	F M F M		N N N N	F F F F		Y Y N Y
Human	M M M M		Y Y Y Y	F M F M		N N N N	F F F F		Y Y N N

Figure 6. Gender, skin tone, and attribute detection results with automated and expert human evaluation. The images are generated by the Stable Diffusion model, using the gender/skin tone-neutral prompts (*e.g.*, "a person who works as a biologist"). For gender estimation, both automated detection and human evaluation agreed on all examples here. For attribute and skin tone estimation, automated detection and human annotation are closely aligned in most cases. The detection results are presented in order of top-left → top-right → bottom-left → bottom-right. *M: Male, F: Female, Y: Yes, N: No.*

Biases in generative image modelling

Stable Bias:
Evaluating Societal Representations in Diffusion Models

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Biases in generative image modelling

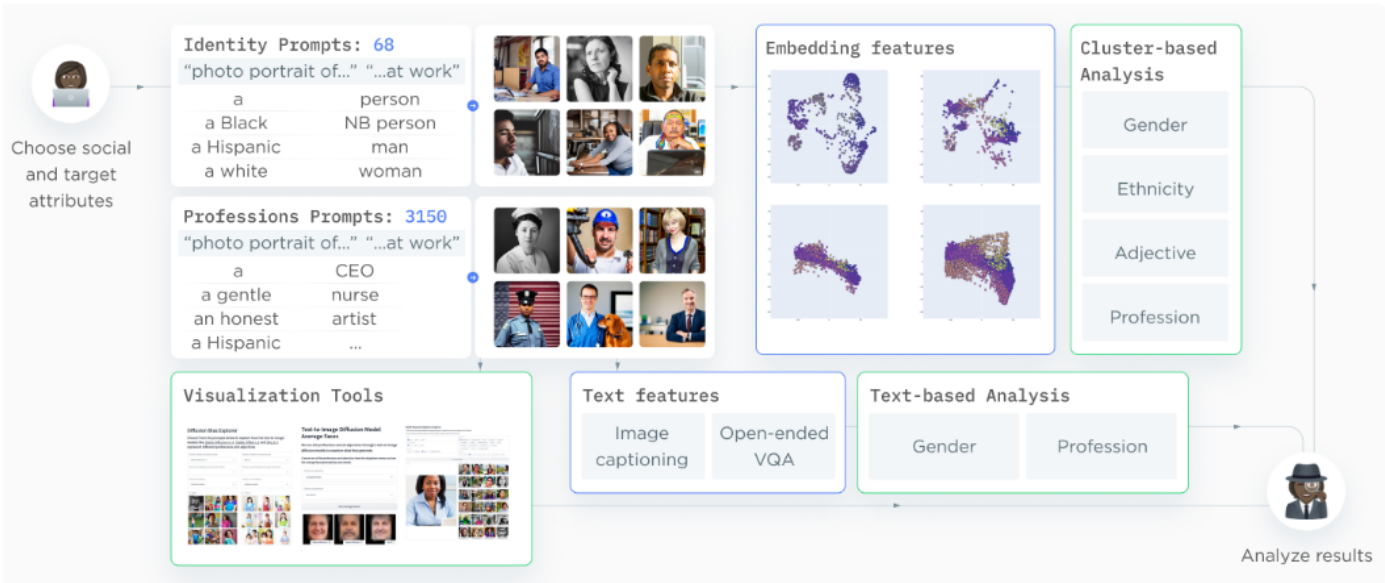


Figure 1: Our approach to evaluating bias in TTI systems

Biases in generative image modelling

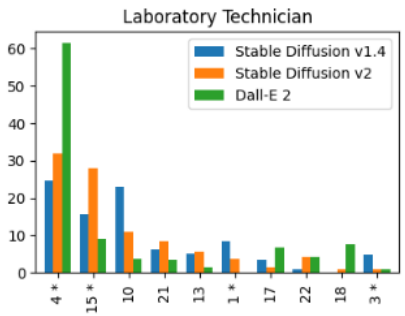


Figure 2: Cluster assignments of "Laboratory technician" images.



Figure 3: Examples of Profession images for "laboratory technician" assigned to clusters 4, 15, 1, and 3. There are no Dall·E 2images assigned to 1.

Biases in generative image modelling

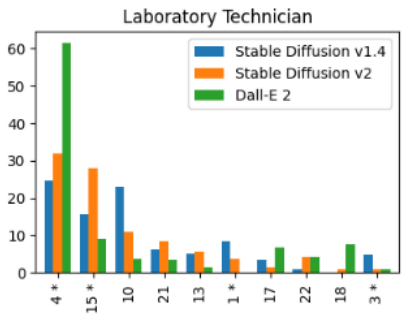


Figure 2: Cluster assignments of "Laboratory technician" images.



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Mitigating Bias

Mitigating biases in text-to-image models: Debiasing text embeddings

Debiasing Vision-Language Models via Biased Prompts

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Mitigating biases in text-to-image models: Debiasing text embeddings

	CelebA		Waterbird	
	Male	Female	Land	Water
$\beta_{y=0}$	0.83	0.78	0.75	0.66
$\beta_{y=1}$	0.77	0.85	0.65	0.70

Table 1: Cosine similarity between classifier weights and spurious directions. In both datasets, the classifier weights are biased toward certain spurious attributes.



Figure 1: Calibration with Positive Pairs. Upon projecting out irrelevant features (such as gender), the embeddings of group prompts should exhibit similarity and contain only information pertaining to the target class (e.g. doctor).

Mitigating biases in text-to-image models: Debiasing text embeddings

4.2 Debiasing via Orthogonal Projection

As the zero-shot weights can also be viewed as natural language embeddings, a straightforward approach is to follow the debiasing pipeline employed in word and sentence embeddings [6, 24]. In particular, to make the classifier invariant to these irrelevant features, we align the classifier weights with the orthogonal complement of these embeddings. Let $A \in \mathbb{R}^{d \times m}$ be a matrix whose columns are the embeddings of spurious prompts. The orthogonal projection matrix is then:

$$P_0 = I - A(A^T A)^{-1} A^T.$$

We can use the projection matrix to eliminate spurious directions in a text embedding z as $P_0 z$.

Mitigating biases in text-to-image models: Debiasing text embeddings



Figure 2: Improving Gender Diversity of Stable Diffusion. We fix the random seed of initial latent noise of Stable Diffusion [36] and generate the images with the training / testing prompt "a photo of a doctor / firefighter". The results demonstrate that applying the calibration matrix to the prompt embedding improves the balance between male and female in the generated images.

Mitigating biases in text-to-image models: Optimizing soft prompts

Published as a conference paper at ICLR 2024

FINETUNING TEXT-TO-IMAGE DIFFUSION MODELS FOR FAIRNESS

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ABSTRACT

The rapid adoption of text-to-image diffusion models in society underscores an urgent need to address their biases. Without interventions, these biases could propagate a skewed worldview and restrict opportunities for minority groups. In this work, we frame fairness as a distributional alignment problem. Our solution consists of two main technical contributions: (1) a distributional alignment loss that steers specific characteristics of the generated images towards a user-defined target distribution, and (2) adjusted direct finetuning of diffusion model's sampling process (adjusted DFT), which leverages an adjusted gradient to directly optimize losses defined on the generated images. Empirically, our method markedly reduces gender, racial, and their intersectional biases for occupational prompts. Gender bias is significantly reduced even when finetuning just five soft tokens. Crucially, our method supports diverse perspectives of fairness beyond absolute equality, which is demonstrated by controlling age to a 75% young and 25% old distribution while simultaneously debiasing gender and race. Finally, our method is scalable: it can debias multiple concepts at once by simply including these prompts in the finetuning data. We share code and various fair diffusion model adaptors at https://sail-sg.github.io/finetune-fair-diffusion/.

Mitigating biases in text-to-image models: Optimizing soft prompts

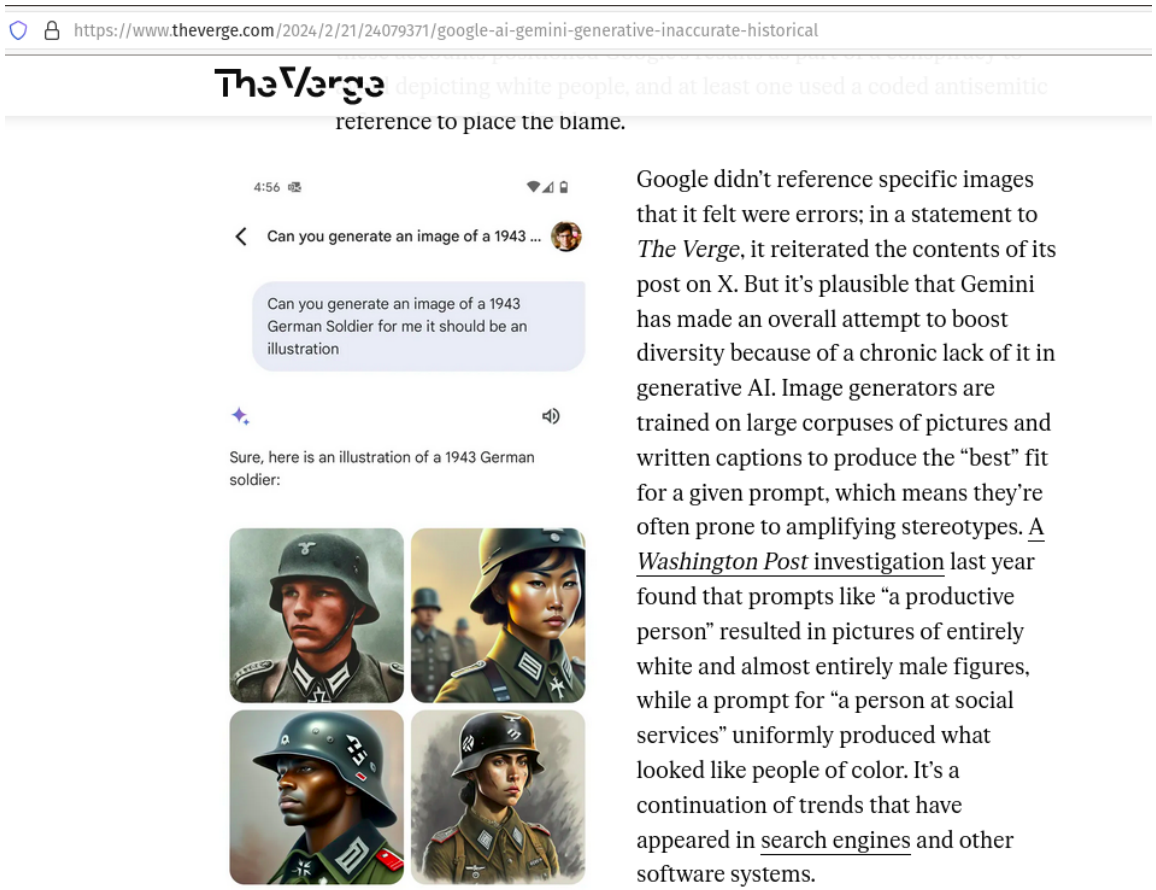
Surprisingly, finetuning as few as five soft tokens as a prompt prefix is able to largely reduces gender bias, demonstrating the effectiveness of soft prompt tuning (Lester et al., 2021; Li & Liang, 2021) for fairness. These results underscore the robustness of our method and the efficacy of debiasing T2I diffusion models by finetuning their language understanding components.



(c) Prompt with unseen context: "bartender at willard intercontinental makes mint julep". Gender bias: 0.87 (original) $\rightarrow$ 0.17 (debiased). Racial bias: 0.49 $\rightarrow$ 0.33. Gender$\times$Race bias: 0.24$\rightarrow$ 0.15.

Mitigating biases in text-to-image models: Optimizing soft prompts

This form of debiasing could backfire though



https://www.theverge.com/2024/2/21/24079371/google-ai-gemini-generative-inaccurate-historical

The Verge

reference to place the blame.

Google didn't reference specific images that it felt were errors; in a statement to *The Verge*, it reiterated the contents of its post on X. But it's plausible that Gemini has made an overall attempt to boost diversity because of a chronic lack of it in generative AI. Image generators are trained on large corpuses of pictures and written captions to produce the "best" fit for a given prompt, which means they're often prone to amplifying stereotypes. A *Washington Post* investigation last year found that prompts like "a productive person" resulted in pictures of entirely white and almost entirely male figures, while a prompt for "a person at social services" uniformly produced what looked like people of color. It's a continuation of trends that have appeared in search engines and other software systems.

Check-in question

What fairness criterion was applied in the Gemini controversy?

Steering through cross attention (ECCV 2026)

EquiSteer: Cross-Attention Steering Towards a Fairer Text-Guided Image Generation

Tatiana Gaintseva[1,2], Akshit Achara[3], Gregory Slabaugh[1], Jiankang Deng[4], and Ismail Elezi[2]

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[2] Huawei Noah's Ark, UK
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Steering through cross attention (ECCV 2026)

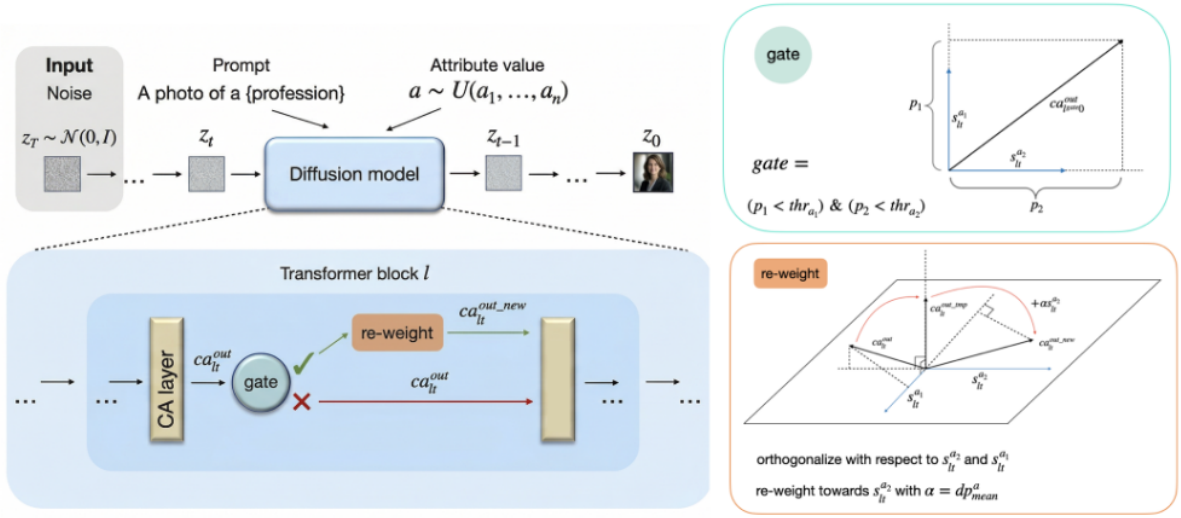


Fig. 2: Illustration of the proposed method, EquiSteer. At each denoising step t and block l, EquiSteer modifies the Cross-Attention (CA) output to promote a desired attribute a. EquiSteer combines three components: (i) a *gate* that estimates attribute presence in the CA output and disables debiasing for attribute-specific prompts, (ii) *orthogonalization* of the CA output w.r.t. $\text{span}(\{s_{lt}^{a_i}\}_{i=1}^{n})$ to remove pre-existing attribute signal, and (iii) *re-weighting* that calibrates the shift magnitude so the target attribute is sufficiently expressed.



(a) Prompt: "A photo of a librarian", SANA **(b)** Prompt: "A photo of a CEO", SDXL

Fig. 6: Qualitative examples of EquiSteer on the *race* concept. Top row in each panel: vanilla model; bottom row: EquiSteer.

Recall the fairness criteria: Which one are we measuring in all we have discussed so far?

- **Independence:** Equal acceptance rates across groups
- **Separation:** Equal TPR, FPR, TNR, FNR across groups
- **Sufficiency:** Equal positive/negative predictive value: Given prediction, equal probability that it is true

Understanding causes of bias

Theoretical results on effect of underrepresentation on generalisation and memorization

THE INTERPLAY OF DATA STRUCTURE AND IMBALANCE IN THE LEARNING DYNAMICS OF DIFFUSION MODELS

A PREPRINT

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[3]Department of Computing Sciences, Bocconi University, Milano, Italy
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[5]School of Computer Science and Applied Mathematics, University of the Witwatersrand, Johannesburg, South Africa

Extreme class imbalance affects dynamics, moderate class imbalance should not

structure dictates the temporal emergence of features.

Result 1 (Class-Specific Speciation Times)**.** *Let b_c denote the sampling weight of class c. The speciation time $t_s^{(c)}$ is governed by two distinct data regimes:*

- ***Weak Imbalance:*** *When all classes are proportionally represented ($b_c = \mathcal{O}(1)$), class imbalance has no leading-order effect on the backward dynamics. All classes speciate simultaneously at:*

$$t_s = \frac{1}{2} \log N + \mathcal{O}(1). \tag{12}$$

- ***Strong Imbalance:*** *When a class is severely undersampled ($b_c = \mathcal{O}(N^{-a_c})$ for $0 < a_c < 1$), its transition is delayed. The system exhibits a hierarchy of class-dependent speciation times:*

$$t_s^{(c)} = \frac{1 - a_c}{2} \log N + \mathcal{O}(1). \tag{13}$$

Properties that affect timescales

Result 2 (Hierarchy of Class Learning)**.** *The learning timescales are governed by the following structural properties, in order of dominance:*

- *Class Variance: High-variance classes present smoother and larger score targets, leading to significantly faster generalization and memorization compared to low-variance components.*
- *Centroid Geometry: Classes with smaller-norm centroids are prioritized over those with large-norm centroids.*
- *Sampling Imbalance: While usually a subleading effect, strong imbalance acts as a modulator capable of inverting the learning order established by variance and geometry.*

Tradeoff between generalization and memorization

Result 3 (The Generalization-Memorization Trade-off)**.** *Sample imbalance impacts generalization and memorization in opposite directions:*

1. *Scarcity delays generalization: Undersampling the high-variance class forces the model to generalize the low-variance class first.*
2. *Scarcity accelerates memorization: Conversely, oversampling the high-variance class deprives the low-variance class of data, causing it to be memorized first.*

Consequently, adjusting sampling weights to close the generalization gap between classes inevitably widens the memorization gap, and vice versa.

Wrapping up

- We are only taking first baby steps towards measuring fairness of generative AI
- Common approaches to quantifying bias: **counting** occurrences of disadvantaged groups or **downstream classification** tasks
- What do we need to improve?
- We cannot expect OpenAI to fix this for us!
- ... so there is still a role – an interesting one – to fill out

Learning goals

By now, you should

- be able to discuss problems and potential solutions in assessing algorithmic fairness of generative AI for computer vision, in particular:
- Measuring bias,
- Mitigating bias, and
- Understanding causes of bias

in diffusion models.

Discussion 1

Dates for oral exam

- Official date is December 10
- Due to co-teachers traveling I cannot examine everyone that date
- Alternative dates could be December 8, 9, 14

Opinions?

Discussion 2

Use of AI in learning/teaching